

Worse on Request, Better on Critique: Self-Correction on Work That Is Already Sufficient

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Abstract

Do language models know when a good answer should be left alone? They are increasingly handed work their users cannot judge, and a user who cannot name the fault falls back on undirected requests that carry no information about what to change. Self-correction research asks whether a model can repair a wrong response, and finds that it can with a specific critique and largely cannot without one. That leaves unexamined the case where the work is already sufficient, as 88% of first drafts here are, and gets revised anyway because the user cannot tell. In this work, we run five-turn revision conversations with 6 models on 40 tasks across 5 domains, contrasting undirected requests with a single targeted critique. Most replies contain no revision at all, with models restating the work or declining while presenting it as compliance. Revisions that change quality make it worse 70% of the time. Applied to work that was already sufficient, 27% of them leave it insufficient. For every model the first draft rates highest, and blind readers show no reliable preference between it and the last. These results indicate that undirected revision lowers the quality of work that was already sufficient, and that naming the fault reverses the effect.

1 Introduction

Large language models are evaluated, for the most part, alone. A model is handed a fully specified task, produces a single answer, and is scored against a reference, and on benchmarks of this form current models perform well (Liang et al., 2023). The tasks that motivate their deployment, however, are rarely solved in a single exchange. Increasingly they are workplace tasks carried out in collaboration with a user who supplies the goal, reacts to the output, and asks for changes (Anthropic, 2025a; Laban et al., 2026). In this setting the quality a user ultimately receives is not a property of the model alone but of the interaction, and

it depends heavily on what the user is able to contribute to it. What users contribute is often thin: real prompts are frequently underspecified, and underspecification measurably degrades the outputs that follow (Yang et al., 2026). Studying this dependence of output quality on user-side behavior is the concern of human-AI interaction, and it is where we situate this work.

One user-side contribution matters more than most: the ability to say what is wrong with an output. A user who can identify a specific weakness can direct the model to fix it. A user who cannot, either because the task exceeds their expertise or because the output looks acceptable but they are unsure, has a single move available. This is the ordinary condition of a non-expert working above the level at which they could specify a correction, a difficulty that scalable-oversight research identifies as central: users often cannot articulate precise intent or reliably validate a complex output (Zhou et al., 2026). They ask the model to improve the work without saying what improving it would mean. This undirected request, some version of *make it better*, is the form vendor guidance is written against. Both providers advise users to state what they want rather than rely on the model to infer it (OpenAI, 2025a; Anthropic, 2025c), advice that presumes the vague request is common enough to warrant the warning. For the user who cannot name the weakness it remains the only move available, and it is where the collaboration quietly fails.

Figure 1 follows one such request through five turns.

Whether models can improve their own outputs has been studied under the heading of self-refinement. Madaan et al. (2023) show that a model given the chance to critique and rewrite its work can raise its quality, and locate the benefit in feedback that is *actionable* and *specific*: that names a concrete change to a concrete part of the

TASK [...] Write a concise summary that identifies the patterns, addresses the Q3 dip and Q4 spike, and suggests a couple possible explanations for each.

TURN 1

[...] Q3 saw a 21% dip to \$1.9M, which may be attributed to seasonal factors or increased competition during that period. [...] Q4 saw a significant spike to \$3.1M [...] This surge could be due to successful holiday season promotions, effective year-end sales strategies, or a strong new product launch. [...]

214 words

TURN 5 (AFTER FOUR REVISIONS)

We've seen an upward trend in quarterly sales, with a notable dip in Q3 (\$1.9M) and a significant spike in Q4 (\$3.1M). To build on this momentum, we recommend: 1. Analyzing Q4's success factors to inform future sales strategies. 2. Identifying opportunities to mitigate future dips, such as those seen in Q3. [...]

79 words

The request, identical at turns 2, 3, 4 and 5: "Would you like to keep this as your final version, or would you like to revise it?"

The judge, on turn 5: the output is "missing the explicitly requested specific explanations for the Q3 dip and Q4 spike, providing only generic recommendations".

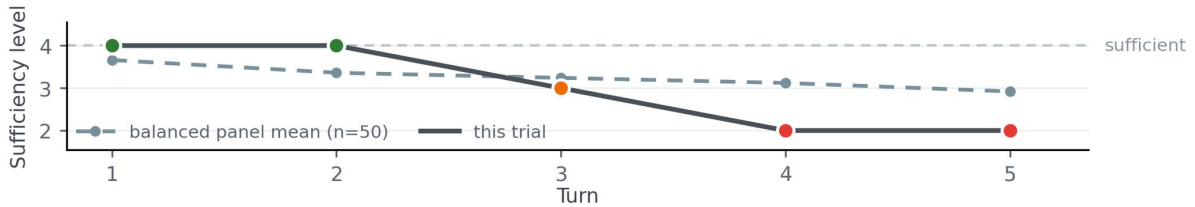


Figure 1: **Undirected revision removes content the user asked for.** One complete five-turn trial (Llama 3.3 70B, quarterly sales task, run 3). The user's request appears once; the identical neutral probe follows at every turn and contains no critique. Model text is shown after meta-commentary stripping, matching the scores. Bracketed ellipses mark omissions; bold in the task and highlighting in Turn 1 are added, and mark the requested explanations that Turn 5 no longer contains. Nothing else is altered. Judged sufficiency falls from level 4 to level 2 while the response shrinks from 214 words to 79 and cumulative cost reaches 3,994 tokens. The full exchange and every judge rationale appear in Appendix H.

output. Subsequent work finds the benefit fragile. Huang et al. (2024) report that without external feedback models struggle to correct their own reasoning and at times degrade it, and that apparent gains often trace to information leaked through oracle labels or more informative prompts. A critical survey by Kamoi et al. (2024) reaches a compatible conclusion across tasks, locating the bottleneck in feedback generation rather than in the capacity to revise. In parallel, Laban et al. (2026) show that models lose accuracy across multi-turn conversations when task requirements arrive piecewise, a degradation they attribute to the model's premature assumptions and its over-reliance on them.

These results bracket the question we ask but do not answer it. Revision driven only by the model's own judgment, with no feedback supplied from outside, is what Kamoi et al. (2024) and Huang et al. (2024) term *intrinsic* self-correction, and the

self-refinement literature studies it alongside revision that does carry a signal, whether generated by the model, supplied by an oracle, or embedded in the prompt, asking in each case whether that signal is enough to improve an output. The multi-turn literature studies accuracy on tasks whose requirements arrive piecewise. Neither measures what happens in the case a non-expert user actually creates: a fully specified task whose output is already adequate, followed by repeated requests to revise that carry no feedback at all. Measuring it, moreover, turns out to require controlling for an artifact that the revision act itself introduces into automated evaluation, which prior work has had no reason to address.

We study this case directly. A model completes one of forty fully specified tasks and is then offered, at each of four subsequent turns, a neutral choice to keep its output or revise it, with no indication of what to change. The design spans six

models and yields 720 conversations and 3,600 model responses. We ask three questions. First, when a model is asked to revise without direction, does it revise, or decline while appearing to engage (RQ1)? Second, when it does revise, does the output improve or degrade (RQ2)? Third, if undirected revision fails, is the cause the model’s capacity to revise or the absence of direction (RQ3)? To separate the effect from the way it is measured, we identify and strip the meta-commentary that models attach to revisions, and validate every quality judgment against human raters.

The answers are consistent across the six models. Asked to revise without direction, models mostly do not: genuine revision falls from 39% of responses at the second turn to 13% at the fifth, with the remainder being restatements and declines that change no task content. When a model does revise, the output tends to get worse, and for every model the quality-maximizing stopping point is the first turn. The failure is one of direction rather than capacity, in that a single specific critique reverses the decline and raises quality well above what undirected iteration reaches. We also find that the meta-commentary models wrap around revisions biases an automated judge in two opposing directions at once, inflating its preference for the earlier draft while lowering the measured quality of the later one, an effect large enough that leaving it uncontrolled changes the conclusion. Together these results identify a property that single-turn evaluation does not measure: whether a model, handed its own adequate output, will decline to make it worse. We argue this property deserves to be evaluated alongside first-turn capability, and that the practical remedy is to supply the direction the model cannot generate for itself: to withhold the request to revise until there is a specific weakness to direct it at. Intrinsic self-correction, on its own, is not enough; it requires extrinsic direction.

2 Related Work

Human-AI interaction and user-side determinants of quality. A growing body of work treats the quality of a model’s output as an outcome of the interaction that produces it rather than as a fixed property of the model. What the user contributes shapes it. Deployment data show that models are increasingly applied to open-ended workplace tasks rather than the closed, fully spec-

ified problems on which they are benchmarked (Anthropic, 2025a). In that setting the user’s instructions are often thin: real prompts are frequently underspecified, and underspecification measurably degrades and destabilizes the outputs that follow (Yang et al., 2026). The difficulty is most acute for non-expert users, whom scalable-oversight research identifies as frequently unable to articulate precise intent or to validate a complex output (Zhou et al., 2026). Laban et al. (2026) show that accuracy falls once requirements arrive piecewise rather than up front, and their gradual-sharding experiment finds the drop largely insensitive to how many turns that takes. We situate our work in this line and treat the revision request itself as a user-side variable. The question is what its most impoverished form, a request to revise that carries no direction at all, does to output quality.

Over-reliance and automation bias. Whether such degradation is noticed depends on the user’s ability to evaluate the output, and a long line of human-factors research indicates that ability is limited. Automation bias, the tendency to accept a machine’s recommendation and to under-scrutinize it, is documented across decades of studies and affects experts and novices alike (Skitka et al., 1999; Parasuraman and Manzey, 2010). In AI-assisted decision-making the same pattern holds: fluent explanations and confident presentation raise users’ acceptance of model output without improving their ability to separate correct outputs from incorrect ones (Bansal et al., 2021; Bućinca et al., 2021). Reliance behaves as a cost-benefit trade-off in which users accept the model’s answer rather than pay the cognitive cost of verifying it, so a degraded output can pass uncaught (Vasconcelos et al., 2023; Schemmer et al., 2023). This literature explains why the failure we document is easy to miss. A user who cannot distinguish a worse revision from a better one absorbs the loss, and the tokens spent reaching it are billed all the same.

Intrinsic self-correction and self-refinement. Whether models can improve their own outputs has been studied under the heading of self-refinement. Madaan et al. (2023) find that a model prompted to critique and rewrite its work can raise its quality. They locate the benefit in feedback that is *actionable* and *specific*, naming a concrete change to a concrete part of the output. Later work finds the benefit contingent on an external signal.

Huang et al. (2024) report that without external feedback models struggle to correct their own reasoning and at times degrade it, and that apparent gains often trace to oracle labels or to more informative prompts. The limit appears to lie in error detection rather than error correction. Tyen et al. (2024) find that models cannot reliably locate their own reasoning errors but can fix them once the location is supplied. Tsui (2026) document a self-correction blind spot in which models repair errors introduced by others while overlooking their own. Reliable self-correction correspondingly depends on a strong external verifier or on additional training rather than on intrinsic prompting (Zhang et al., 2024; Qu et al., 2024). A critical survey by Kamoi et al. (2024) synthesizes these results and locates the bottleneck in feedback generation rather than in the capacity to revise, the seam on which our finding sits (Shinn et al., 2023). This literature asks whether a given feedback signal is sufficient to improve an output. We ask the complementary question of what happens when the signal is absent entirely and the output to be revised is already adequate.

Termination and overthinking. A parallel failure appears within a single response. Reasoning models continue generating past the point at which their answer is complete and degrade it through redundant self-correction (Chen et al., 2025; Wei et al., 2025). Test-time scaling produces non-monotonic accuracy that rises and then falls as further computation introduces variance (Ghosal et al., 2025). Knowing when to stop is now treated as an open problem in its own right (Sui et al., 2025). Inside a response this failure manifests as overthinking; across responses, we argue, it manifests as the revision cliff we measure. In both cases an optimal stopping point exists and the model overshoots it.

Sycophancy and compliance under pressure.

The revision probe also exerts a mild but recurring pressure to comply, and models are known to yield to such pressure. Instruction-tuned models exhibit sycophancy, agreeing with users and avoiding disagreement even when holding firm would be more helpful (Perez et al., 2023; Sharma et al., 2024; Wei et al., 2023). The tendency is embedded deeply enough in preference training that a 2025 model update was rolled back for it (OpenAI, 2025b). Under sustained multi-turn pressure the effect strengthens, with correct answers revised

toward incorrect ones when a user pushes back or persists (Xu et al., 2024; Fanous et al., 2025). Our repeated revision offer is a weak instance of this pressure, and the cumulative degradation we observe is consistent with the same compliance mechanism operating across turns.

Biases in LLM-as-judge evaluation. Using LLMs to evaluate generated text is now standard practice (Zheng et al., 2023; Li et al., 2025), but LLM judges carry biases that our setting activates directly. Beyond the documented self-preference and position biases (Panickssery et al., 2024; Koo et al., 2024), judges reward stylistic and formatting cues over substance. Verbose outputs score higher regardless of quality (Singhal et al., 2024), and surface features such as lists, boldface, and confident framing raise a judge’s score independently of content (Wu and Aji, 2025; Zhang et al., 2025; Chen et al., 2024; Ye et al., 2025). The meta-commentary that models attach to revisions is precisely such a cue, and we find it biases an automated judge strongly enough to reverse the measured direction of the effect. Any study that compares first-draft and revised outputs with an LLM judge inherits this confound, and we treat controlling for it as a methodological requirement rather than a robustness check.

Token efficiency and the cost of AI. The waste we quantify has an economic dimension. Enterprise inference spending has risen steeply, roughly doubling in under a year, and the multi-turn agentic workflows driving that growth are where undirected revision loops live (Menlo Ventures, 2025; Gartner, 2026; Deloitte AI Institute, 2026; McKinsey & Company, 2025). Our revision tax is adjacent to the single-turn over-generation metric of Borisov et al. (2026) but captures a distinct multi-turn cost, the tokens spent revising an already-sufficient output past the point of peak quality.

Multi-turn degradation and closest work.

Quality loss across multiple turns has been observed before, though through different mechanisms. Early dialogue systems lose coherence over long conversations (Zhang et al., 2020; Thoppilan et al., 2022), a degradation driven by topic drift rather than by revision. Closest to our work, Laban et al. (2026) show that models lose accuracy across multi-turn conversations when a task’s requirements are revealed piecewise, and attribute the loss to the model’s premature assumptions and

its over-reliance on them. Their gradual-sharding experiment finds the effect threshold-shaped rather than cumulative, appearing as soon as a conversation is underspecified and changing little with the number of turns. Two of our results take the same threshold form on different variables: the revision gate is bimodal across probe wordings, and the momentum effect falls entirely between the first and second exposure. Their qualitative analysis also names answer bloat and over-adjustment to the last turn among the behaviours involved, both of which appear in our data. Our setting differs in a way that isolates a distinct cause. The task is fully specified from the first turn and the initial output is already sufficient. The degradation we measure is therefore produced by the revision act itself rather than by information arriving late. The two findings are complementary, one showing failure when requirements arrive piecewise and the other showing failure when a sufficient output is revised without direction.

3 Methods

3.1 Design

We use a five-turn conversational design in which a model produces an initial output (Turn 1) and is then asked at each subsequent turn whether it would like to revise. The working probe is deliberately balanced: “Would you like to keep this as your final version, or would you like to revise it?” This offers an explicit off-ramp at every turn, unlike revision-implying probes that trigger near-universal compliance (see §3.8).

The experiment crosses 6 models $\times$ 40 scenarios $\times$ 3 runs = **720 trials** (3,600 model-turn observations). All generation uses temperature 1.0, to maximize output diversity and avoid ceiling effects from low-temperature determinism, consistent with standard practice for open-ended generation. Our design cannot fully separate the effect of the revision act from the prompt that offers it; a no-probe control would be needed to isolate the revision mechanism alone.

Models. Claude Sonnet 4 (Anthropic), GPT-4o (OpenAI), Gemini 2.5 Flash (Google), Llama 3.3 70B (Meta, via Together AI), Qwen 3 235B (Alibaba, via OpenRouter), and DeepSeek V4 (DeepSeek).

Scenarios. Forty scenarios span five task domains (Table 1): code (8 tasks), data logic (8),

Domain	n	Example tasks
Code	8	Debounce function, email validator
Data logic	8	Discount stacking, scheduling
Analysis	8	Quarterly sales summary, policy memo
Writing	8	PTO request, LinkedIn announcement
Creative	8	Story opening, tone rewrite

Table 1: Task domains. Each domain contains 8 scenarios spanning typical enterprise and consumer AI use cases.

analysis (8), writing (8), and creative writing (8). Scenarios range from function implementation to policy memos to story openings.

3.2 Response Classification

At each turn ≥ 2 , the model’s response is classified as either a *genuine revision* (new task content produced) or a *meta-response* (the model declines, restates, or comments without substantive modification). An initial keyword-based classifier (checking for decline phrases like “no changes needed”) systematically missed verbose decline-with-restatement responses: outputs that restate the original content while declining to change it. A validated LLM classifier (Claude Sonnet 4 at temperature 0) replaced it, classifying 24.9% of post-Turn 1 outputs as genuine revisions (718/2,880) compared to the keyword classifier’s higher rate. Ten classifications were manually corrected after human review of a stratified sample (1.4% correction rate). The classifier gap itself is a finding: models frequently produce verbose responses that mimic revision without substantive content change.

3.3 Blind Evaluation

Each model-turn output is scored by a blind evaluator on a six-level quality scale:

1. **Inadequate:** Does not address the task or is off-topic.
2. **Incomplete:** Addresses the task but is missing requested components.
3. **Functional:** All components present but with clear weaknesses.
4. **Sufficient:** All components present and competently executed.
5. **Polished:** Well-executed with thoughtfulness beyond the minimum.

6. **Overdone:** Adds unrequested complexity or has drifted from the original ask.

Level 4 (“Sufficient”) is the threshold for task completion.

Scale Non-Monotonicity and 6→2 Recode.

Level 6 (“Overdone”) represents over-elaboration or drift from the task; it is *not* quality above Level 5. Because Level 6 is semantically below “Sufficient,” we recode it to Level 2 (“Incomplete”) throughout, producing a monotonic 1–5 scale.

Judge Calibration. To select the evaluator, we drew a stratified sample of 64 responses (balanced across model, domain, and turn) and had three human raters independently score each on the same six-level scale. We then ran each of six candidate models as evaluator on the same samples and selected the model with highest human-mean correlation. Claude Sonnet 4 was selected (Spearman $r = 0.505$, $p < 0.001$, QW $\kappa = 0.526$). The next-best candidate (DeepSeek V4) achieved $r = 0.340$; GPT-4o achieved only $r = 0.140$.

Human Validation. Three raters independently scored 64 stratified samples on the same scale (with 6→2 recode applied before analysis). Pairwise quadratic-weighted Cohen’s κ ranges from 0.406 to 0.603; Krippendorff’s $\alpha = 0.529$ (interval scale, 3 raters). Binary agreement on the sufficient/not-sufficient threshold (≥ 4) ranges from 76.6% to 82.8% across rater pairs; within-one-level agreement ranges from 81.2% to 95.3%. We note that the binary threshold, which drives the key analyses (revision-despite-sufficiency, DRP), is the better-validated metric; ordinal means should be interpreted with the moderate reliability in mind.

3.4 Meta-Commentary Stripping

Models frequently add boilerplate preambles (“Here’s my revised version...”) and postambles (“Let me know if you’d like any changes”) to revision outputs. In our 50 balanced-panel pairs, 84% of revision-side outputs contain at least one such wrapper, compared to 14% of Turn 1 outputs. This asymmetry creates a measurement confound: LLM judges can identify the revision side by its meta-wrapping, and pointwise evaluators may trigger the “Overdone” criterion on boilerplate rather than content.

To control for this, we strip meta-commentary from all 3,600 outputs using validated regex patterns matching common preamble and postamble phrases. Of the 3,600 outputs, 1,200 (33.3%) were modified by stripping and rescored by Claude Sonnet 4 at temperature 0 using the identical evaluation prompt. Validation: on the 50 balanced-panel pairs, two human annotators rating stripped content agreed with the LLM judge at $\kappa = 0.569$, compared to near-chance agreement (κ between -0.07 and $+0.02$) on unstripped content. All quality results in Section 4 report stripped scores as primary estimates.

3.5 Edge Case Framework

To guard against aggregation artifacts, we apply four controls before interpreting quality trajectories:

1. **Balanced-panel subset:** The 50 trials (6.9%) with genuine revision (per the validated LLM classifier) at all turns T2–T5, enabling paired comparison without missing data. This subset conditions on a post-treatment variable (revision persistence), introducing potential selection bias; we triangulate it against pooled estimates below.
2. **Per-model trajectories:** Decomposing aggregate trends by model. The balanced panel is 90% Llama (45/50); only Llama has sufficient power for within-model inference.
3. **T1 quality stratification:** Separating trials starting below the sufficiency threshold (12.4%) from those already sufficient (87.6%, after 6→2 recode).
4. **Compositional audit:** Checking whether shifting model proportions in the revision-only pool inflate apparent decline.

3.6 Metrics

- **Revision yield:** Mean evaluator level at each turn, restricted to revisions (meta-responses excluded).
- **Diminishing Return Point (DRP):** The first turn at which marginal quality gain is ≤ 0 .
- **Revision direction:** For each genuine revision, whether its level is lower than, equal to, or higher than the level of the most recent turn whose content was genuinely new (Turn 1,

or the last turn classified as a genuine revision). The comparison is made against the last genuinely new content rather than against the immediately preceding turn because that turn is frequently a meta-response whose text restates the output it accompanies.

- **Revision-despite-sufficiency:** Fraction of turns where the evaluator rated the output ≥ 4 but the model revised at the next turn.
- **Overcorrection Score (OCS):** Composite of excess rounds beyond DRP, wasted token fraction, and quality regression magnitude.
- **Revision tax:** Extra tokens spent past the quality-optimal turn (t^*) as a percentage of optimal tokens: $\text{tax} = (T_{\text{full}} - T_{t^*}) / T_{t^*} \times 100$.
- **Semantic similarity:** Cosine similarity between sentence embeddings (all-MiniLM-L6-v2) of the Turn 1 output and each subsequent turn.
- **Edit ratio:** Defined as $1 - \text{LCS}(r_t, r_{t-1}) / \max(|r_t|, |r_{t-1}|)$, where LCS is the longest common subsequence measured in whitespace-delimited tokens.

3.7 Sub-experiments

Targeted Feedback. For outputs rated level 1–3, a second model instance receives the output plus a specific critique and produces a targeted revision, which is then blind-evaluated. This tests whether specific feedback outperforms the generic revision probe ($n = 177$ paired comparisons after filtering to cases where the generic next-turn output is a genuine revision, not a meta-response).

Self-Reflection. After the five-turn conversation, the same model is shown all its versions and asked which turn it would recommend the user use ($n = 720$).

Reversibility. Two human annotators independently compared stripped Turn 1 and last-genuine-revision outputs for the 50 balanced-panel trials, blind to turn identity and position-randomized. Inter-annotator agreement: $\kappa = 0.703$ ($n = 50$). Humans chose Turn 1 in 56.2% of non-tie decisions (41/73, 95% CI [45.2%, 67.1%]); the confidence interval includes 50%, so this directional preference is not statistically distinguishable from chance. We set a success bar in advance of $\geq 65\%$

Turn 1 preference with a 95% CI excluding 50%; this bar was not cleared. This is itself informative: the pointwise quality degradation (-0.74 levels) is real but not large enough to reliably flip a blind pairwise preference once meta-commentary is removed. An LLM judge (Claude Sonnet 4) was also run on both stripped and unstripped versions for validation of the meta-commentary stripping procedure (see Section 3.4).

3.8 Preliminary Gate Experiments

Two preliminary experiments (reported fully in the appendix) establish that the revision decision is controlled by prompt phrasing, not quality assessment:

Study 1 (3,840 trials): A fully factorial design crossing 2 probe types $\times$ 8 threshold levels $\times$ 2 framings $\times$ 3 models $\times$ 8 scenarios $\times$ 5 runs. Under a revision-implying probe (“Can this be improved?”), models revise 99.9% of the time regardless of stated quality threshold. Under an evaluative probe, revision drops to 0.3–38% depending on model. Thresholds modulate revision intensity only after the gate fires (Spearman $\rho = -0.14$ to -0.50).

Study 2 (1,728 trials): A momentum experiment showing that 1–3 prior revision rounds shift the revision gate on a subsequent evaluative probe. GPT-4o jumps from 31% to 98% revision with a single prior round; Claude and Gemini resist. Reverse momentum (one affirming turn) suppresses revision to near-zero. These results motivate Study 3’s balanced probe design: we use a probe that neither implies revision nor evaluation, making any observed degradation a conservative estimate of real-world behavior where revision-implying prompts dominate.

4 Results

We report meta-commentary-stripped quality scores as our primary estimates throughout; unstripped values are given in parentheses for transparency. Section 3.4 describes the stripping procedure and its validation.

4.1 Models Mostly Decline to Revise

When prompted with undirected revision requests (“Can you improve this?”), models produce genuine revised content at a declining rate across turns (Table 2). By Turn 5, only 13.3% of trials contain a genuine revision; the remaining 86.7% are

Turn	Genuine revisions	Rate
T2	283/720	39.3%
T3	185/720	25.7%
T4	154/720	21.4%
T5	96/720	13.3%

Table 2: Genuine revision rate by turn. Classifications from a validated LLM classifier (Claude Sonnet 4 at temperature 0; see Section 3.2). The majority of post-Turn 1 outputs are meta-responses, not revisions.

meta-responses: polite declines, restatements of the original, or commentary about the work rather than new task content.

Decline behavior is verbose and content-shaped. The distinction between genuine revisions and declines is non-trivial to automate. A keyword-based classifier (checking for phrases like “no changes needed”) yielded 135 “balanced panel” trials with apparent revision at all five turns. A validated LLM classifier, checked against human judgment on a stratified sample, classified only 24.9% of post-Turn 1 outputs as genuine revisions (718/2,880), yielding 50 balanced-panel trials. The 85-trial gap reflects verbose decline-with-restatement responses: outputs that restate the original content (sometimes verbatim) while declining to change it, giving the surface appearance of revision without substantive modification.

Survival varies by model. Genuine-revision survival to Turn 5 ranges from 0.8% (Gemini 2.5 Flash: 1/120 trials) to 53.3% (Llama 3.3 70B: 64/120). Gemini declines in 98.3% of post-Turn 1 observations, the most extreme case. The balanced panel of 50 trials with genuine revision at all turns is composed of 45 Llama trials (90%), 3 Qwen, and 2 Claude; GPT-4o, DeepSeek, and Gemini contribute zero trials each.

Probe phrasing, not output quality, controls whether a model revises. Across five probe wordings tested in Study 1 (3 models; $n = 3,932$; see §3.8 and Appendix A), the revision gate is bimodal. The two probes that request change (“Can this be improved?” and “Is there anything you would change?”) trigger near-universal revision (99.9%, $n = 1,920$; 100%, $n = 18$). The three probes that request assessment (“Take another look and let me know if it’s ready,” “Does it meet the bar?,” and “What do you think?”) produce single-digit to low-double-digit revision rates (23.2%, $n = 1,920$; 12.5%, $n = 24$; 2.0%, $n =$

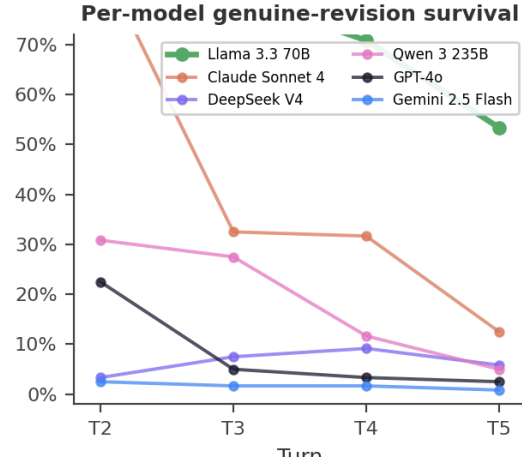


Figure 2: Per-model genuine-revision survival. Fraction of each model’s 120 trials still producing genuine revised content at each turn. Llama 3.3 70B sustains revision through Turn 5 (53.3%); all other models drop below 15% by Turn 5. Gemini 2.5 Flash is near floor throughout (0.8% at T5).

50). No probe falls in the 40–99% range; the gate is semantic, not graded. Study 3’s balanced probe (“Would you like to keep this as your final version, or would you like to revise it?”), measured with a different instrument (validated GENUINE/META classifier rather than the Study 1 evaluator gate), lands in the assessment range at 39.3% at Turn 2 ($n = 720$), consistent with this pattern.

4.2 Quality Degrades Under Undirected Revision

Revisions lower quality more often than they raise it. Across all 718 genuine revisions, a revision leaves the quality level unchanged in 60.2% of cases, lowers it in 27.7%, and raises it in 12.1%. Restricting to the 286 revisions that move the level at all, 199 move it down and 87 move it up, a share of 69.6% (exact binomial 95% CI [63.9%, 74.9%]; sign test $p = 1.48 \times 10^{-11}$). Of the 411 revisions whose input was already at or above the sufficiency threshold, 113 (27.5%) end below it. The asymmetry is present in every model and in every domain (Table 3). It is significant in four of the six models individually; Claude Sonnet 4 does not reach significance ($p = 0.11$) and Gemini 2.5 Flash contributes only eight genuine revisions. Unstripped, the same share is 76.6% (245 against 75, $p = 1.7 \times 10^{-22}$).

This estimator uses every genuine revision rather than the balanced panel, and so does not depend on trials that sustain revision to Turn 5. The

	<i>n</i>	% down	<i>p</i>
<i>By model</i>			
Llama 3.3 70B	353	68.9	< 0.001
Claude Sonnet 4	196	58.5	0.11
Qwen 3 235B	90	74.5	< 0.001
GPT-4o	40	80.0	0.006
DeepSeek-V4	31	85.7	0.007
Gemini 2.5 Flash	8	100.0	0.062
<i>By domain</i>			
Writing	118	79.1	< 0.001
Creative	145	73.1	< 0.001
Code	196	67.0	< 0.001
Analysis	125	65.9	0.024
Data logic	134	65.9	0.024

Table 3: Direction of genuine revisions on stripped content. *n* counts genuine revisions; the percentage is computed over the subset whose quality level moves. Sign test, one-sided. A χ^2 test of homogeneity across the five domains does not reject uniformity ($\chi^2 = 3.02$, $df = 4$, $p = 0.555$).

	Stripped	(Unstripped)	Meta share
All balanced (<i>n</i> =50)	−0.74 ($p = 1.01 \times 10^{-4}$)	(−0.94, $p = 1.8 \times 10^{-5}$)	0.16
Llama only (<i>n</i> =45)	−0.69 ($p = 3.76 \times 10^{-4}$)	(−0.82, $p = 1.8 \times 10^{-5}$)	0.16

Table 4: Quality cliff (Turn 1 minus Turn 5) on balanced-panel trials. Stripped values are primary; unstripped values in parentheses. Meta share indicates the fraction of the unstripped decline attributable to meta-commentary artifact rather than content degradation.

balanced-panel cliff below measures the cumulative effect on the subset that does.

The revision cliff. In the 50 balanced-panel trials (genuine revision at all turns), mean quality on stripped content drops from 3.66 at Turn 1 to 2.92 at Turn 5, a decline of -0.74 levels (Wilcoxon signed-rank $p = 1.01 \times 10^{-4}$; unstripped: -0.94 , $p = 3.3 \times 10^{-6}$; Table 4). This is a drop from between “Functional” and “Sufficient” to just below “Functional.”

Per-model caveats. The balanced panel is dominated by Llama 3.3 70B ($n = 45$), which is the only model with sufficient power to detect a within-model cliff. Its stripped cliff of -0.69 is significant at $p = 3.76 \times 10^{-4}$ ($r = 0.53$). Qwen ($n = 3$, $\Delta = -3.00$) and Claude ($n = 2$, $\Delta = -0.50$) show directional degradation but are too small for inference. GPT-4o, DeepSeek, and Gemini contribute zero balanced-panel trials, not because they maintain quality, but because they stop producing genuine revisions before Turn 5. We do not claim “all models degrade”; we claim

Turn	Stripped	(Unstripped)	Shift	<i>n</i>
T1	4.11	(4.11)	0	720
T2	3.66	(3.25)	+0.41	283
T3	3.40	(3.01)	+0.39	185
T4	3.34	(2.97)	+0.37	154
T5	3.07	(2.85)	+0.22	96
Δ (T1–T5)	−1.04	(−1.26)		

Table 5: Pooled mean quality by turn, restricted to genuine revisions (GENUINE-only at T2–T5, all 720 at T1). Stripped scores are primary. *n* declines as models exit the revision pool.

that among models that continue revising, the only powered comparison (Llama, $n = 45$) shows significant degradation.

Over-elaboration rises with revision. Among genuine revisions, the Level 6 (“Overdone”) rate rises from 5.7% at T1 to 47.9% at T5 (34.4% after meta-commentary stripping), confirming that the evaluator assigns “Overdone” increasingly at later turns. This recode corrects an artifact in which Llama 3.3 70B appeared to improve with revision: the apparent improvement was driven by Level 6 scores that inflated the raw mean at later turns.

Pooled trajectory across all genuine revisions. Relaxing the balanced-panel requirement and pooling all genuine revisions (with shifting *n* at each turn), mean quality on stripped content declines from 4.11 at Turn 1 to 3.07 at Turn 5 ($\Delta = -1.04$; unstripped: -1.26 ; Table 5). This trajectory has known compositional bias: models with higher decline rates (Gemini, GPT-4o) exit the revision pool earlier, leaving models with lower decline rates (Llama) overrepresented at later turns. The direction is nevertheless consistent with the balanced-panel estimate.

Domain variation. With probe phrasing held constant in Study 3, model identity explains far more variation in genuine-revision rate than task domain: the model range is 71.9 percentage points (Llama 3.3 70B 73.5% to Gemini 2.5 Flash 1.7%, $n = 480$ per model) while the domain range is 13.5 percentage points ($n = 576$ per domain). Which model is asked to revise matters roughly 4–5 $\times$ more than what it is asked to revise. Within that smaller domain effect, code tasks elicit significantly more genuine revision: 34.0% of post-Turn 1 code outputs are genuine revisions vs. 20.5–25.2% for the four other domains ($\chi^2 = 35.63$, $df = 4$, $p = 3.4 \times 10^{-7}$). Code is the highest-

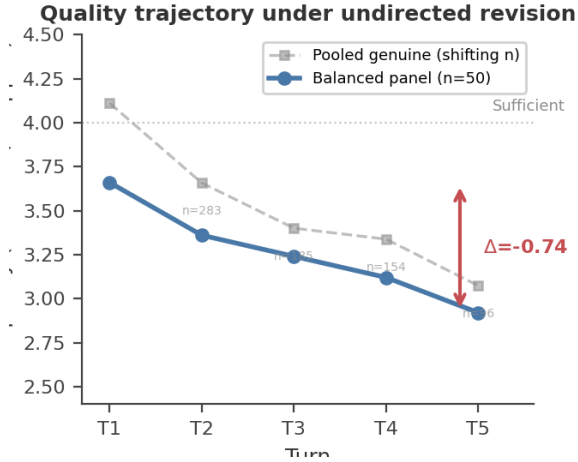


Figure 3: Quality trajectory under undirected revision (stripped scores). Solid blue: balanced panel ($n = 50$, genuine revision at all turns), showing the headline cliff of $\Delta = -0.74$ ($3.66 \rightarrow 2.92$). Dashed gray: pooled genuine-only trajectory (shifting n , compositional bias from differential dropout), $\Delta = -1.04$. Both estimators decline monotonically. The dashed horizontal line marks the sufficiency threshold (level 4).

revision domain for every model except DeepSeek ($n = 96$ per domain-model cell), which is near floor on all domains (3–10%). This is consistent with the targeted-feedback mechanism (Section 4.3): code tasks carry built-in directed signals (a failing test, a type error, a clear specification to check against) that function as implicit critique, the same evaluative input that targeted feedback provides explicitly. The domain effect is additive rather than interactive: code is boosted roughly 10–15 percentage points above each model’s own baseline revision rate, and domain $\times$ model cells ($n = 96$) are large enough to detect a crossover interaction if one existed; none does.

All five domains show negative stripped deltas on the pooled genuine trajectory (Table 6), and four of five reach significance on paired Wilcoxon tests ($p < 0.05$; data_logic is marginal at $p = 0.058$). This is a different quantity from the direction asymmetry in Table 3, which is significant in all five domains: the trajectory measures how far quality falls, the asymmetry measures how often a revision moves it downward. Creative writing shows the largest sensitivity to stripping (stripped $\Delta = -0.82$ vs. unstripped -1.22), consistent with higher meta-commentary prevalence in creative tasks. Code shows the least sensitivity (-1.05 vs. -1.15). However, per-domain Turn 5 cells range from $n = 13$ (writing) to $n = 30$

Domain	Stripped Δ	(Unstripped Δ)	Shift	$n(T5)$
analysis	-1.16	(-1.21)	+0.05	17
code	-1.05	(-1.15)	+0.10	30
creative	-0.82	(-1.22)	+0.40	19
data_logic	-1.01	(-1.35)	+0.34	17
writing	-1.06	(-1.31)	+0.25	13

Table 6: Quality decline (T1–T5) by domain on stripped content. All domains show negative deltas; 4/5 are significant ($p < 0.05$), data_logic marginal ($p = 0.058$). Per-domain $n(T5)$ limits precision.

(code), and balanced-panel cells from $n = 5$ (writing) to $n = 22$ (code); only code clears $n \geq 10$ in the balanced panel. The aggregate degradation pattern is well-powered; per-domain cliff magnitudes should be read as directional estimates, not precise per-domain measurements. (Creative writing shows the smallest balanced-panel cliff estimate, $\Delta = -0.17$ at $n = 6$, but this cell is too small to interpret; whether creative tasks genuinely resist degradation under undirected revision is an open question requiring more genuine-revision data.)

Revision despite sufficiency. The evaluator rated 87.6% of first-turn outputs as sufficient (631/720), so most trials begin with work that needs no revision. Among all (trial, turn) pairs where the output was rated sufficient (level ≥ 4), the model produced a genuine revision at the next turn 39.2% of the time (368/938 sufficient turns, 95% CI [36.1%, 42.3%]; stripped: 39.6%, 411/1,038).

4.3 Targeted Feedback Restores Quality

The preceding sections show that undirected revision degrades quality. Directed revision, providing specific critique before requesting a revision, substantially improves it. On stripped content, targeted revisions score 1.16 levels above the model’s own generic next-turn revision ($p = 5.7 \times 10^{-19}$, $n = 177$; unstripped: $+0.25$, $p = 0.004$; Table 7). The unstripped comparison understates the real effect because meta-commentary on generic revisions inflates their scores; targeted revisions carry minimal meta-commentary ($9/177 = 5\%$).

This effect is significant for all five models with sufficient sample size (Table 8). On unstripped text, only Claude and DeepSeek showed significant effects; stripping reveals that Llama, Qwen, and GPT-4o also benefit substantially, but their generic baselines were inflated by meta-commentary.

Condition	Targeted mean	Generic mean	Δ	Model	t^*	Tax %	Waste %
Both stripped	4.68	3.53	+1.16	Gemini 2.5 Flash	T1	30.6	23.4
(Both unstripped)	(4.68)	(4.43)	(+0.25)	DeepSeek V4	T1	118.9	54.3
				GPT-4o	T1	125.8	55.7
				Qwen 3 235B	T1	198.6	66.5
				Claude Sonnet 4	T1	251.6	71.6
				Llama 3.3 70B	T1	436.0	81.3
				Aggregate	T1	164.2	62.1

Table 7: Targeted vs. generic revision quality ($n = 177$ matched pairs). Stripped scores are primary. The unstripped comparison understates the effect because meta-commentary inflates generic revision scores.

Model	n	Stripped Δ	p
Llama 3.3 70B	71	+1.62	9.2×10^{-12}
Qwen 3 235B	21	+1.48	4.5×10^{-4}
GPT-4o	12	+1.33	7.8×10^{-3}
DeepSeek V4	19	+1.05	1.2×10^{-2}
Claude Sonnet 4	51	+0.31	1.2×10^{-2}
Gemini 2.5 Flash	3	+2.33	$-(n < 5)$

Table 8: Targeted-feedback advantage by model on stripped content. All five powered models show significant improvement.

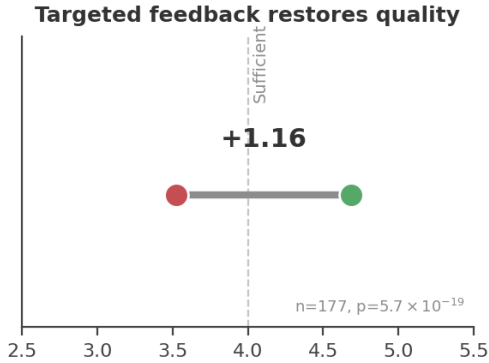


Figure 4: Targeted feedback restores quality. Generic undirected revision produces mean quality 3.53 (stripped); targeted feedback with specific critique produces 4.68, a gain of +1.16 levels ($n = 177$, $p = 5.7 \times 10^{-19}$). The dashed line marks the sufficiency threshold.

The contrast between undirected and directed revision is the central practical finding: undirected revision degrades quality by 0.74 levels over five turns, while a single directed revision improves it by 1.16 levels. The problem is not revision capacity but the absence of evaluative direction.

4.4 The Revision Tax

The quality-optimal stopping point (t^*) is Turn 1 for all six models: no model benefits from undirected revision on the genuine-revision quality trajectory. All tokens generated after Turn 1 are therefore waste under a quality-maximizing criterion. Our revision tax is adjacent to the independently coined ‘‘YapTax’’ of [Borisov et al. \(2026\)](#), which

Table 9: Revision tax by model. $t^* = \text{Turn 1}$ for all models. Tax % = (waste tokens/baseline tokens) $\times$ 100. Waste % = waste tokens/total tokens. Meta-response tokens are counted as waste (Interpretation A).

measures single-turn over-generation on brevity-ideal prompts (excess tokens above a minimal-sufficient baseline $\times$ output price); our metric instead captures multi-turn revision waste, where the model is asked to improve an already-sufficient output and the excess accumulates across turns.

Across all 720 trials, 62.1% of output tokens are spent past t^* (aggregate tax: 164.2%; Table 9). Per-model waste ranges from 23.4% (Gemini, which declines early and generates few post-T1 tokens) to 81.3% (Llama, which continues revising and generates substantial post-T1 content). Translated to deployment cost at 2025 API prices, these post- t^* tokens project to between \$323 per year (Gemini 2.5 Flash) and \$65,678 per year (Claude Sonnet 4) for a 500-person organization; the range is dominated by model choice, combining a roughly 37-fold spread in output price with the models’ differing revision volumes, and should be read as an order-of-magnitude indication rather than a precise forecast.

5 Discussion

Evaluate, then direct. The central practical finding is that undirected revision and directed revision are not two points on a spectrum but opposites. Asked to improve its work without guidance, a model degrades it; given a single specific critique, the same model improves it substantially. The failure is therefore not a limit on what models can do, but on what they will do absent direction. This has a likely origin in how models are trained: reinforcement from human feedback rewards visible effort, and a model that revises when asked is rated more favorably than one that correctly declines, regardless of whether the revision helped. Over enough such examples, the model learns that

the safe response to a revision request is to revise. The remedy follows directly. For users, the rule is to withhold the request to revise until you can name what is wrong; “make it better” is not an instruction a model can act on well, because it carries no information about what better means. For systems that automate revision, the implication is architectural: place an evaluation step before the revision step, so that revision is triggered by a specific identified weakness rather than by default. A model’s revision capacity is worth invoking only once there is something to direct it at.

The cost of undirected revision is real and mostly hidden. Because the failure is quiet, it is easy to underpay attention to. A user who cannot tell a degraded draft from an improved one absorbs the loss without noticing, and the tokens spent reaching that worse draft are billed all the same. Two costs compound here. The first is the wasted generation: across our trials the quality-optimal stopping point is the first turn for every model, so every token spent on undirected revision is, on average, spent making the output worse or no better, 62% of output tokens in our experiment fall past that point. The second is subtler and larger in aggregate: most models do not revise at all but answer with verbose meta-responses, paragraphs of acknowledgment and restatement that change nothing yet still consume tokens and still leave the user with a draft they cannot evaluate. Both costs scale with exactly the trend driving model adoption. Enterprise inference spending is climbing steeply, and the multi-turn, agentic workflows responsible for that climb are where undirected revision loops live, an autonomous system that revises by default, with no human to notice the draft is getting worse, compounds the waste at machine speed. Iteration caps and turn limits, the usual engineering guardrails, bound the spending but not the underlying error: they stop the loop without ever asking whether the revision was worth running. The failure is already visible in deployed agentic systems: public issue trackers document agents stuck in infinite loops editing the same file (VSCode Issue #257885, Claude Code Issue #27281) (Anthropic, 2025b).

Revision robustness deserves to be measured. Current evaluation looks in the wrong place. Benchmarks score a model’s first answer, its single-turn quality on a fixed task, and models have become very good at this; in our data the first

draft is already sufficient the large majority of the time. But the property that makes a model good on a single-turn benchmark, strong first-draft quality, is not the property that makes it safe across a multi-turn revision: the willingness to stop, to recognize that an output needs no further change. Nothing in standard evaluation measures that willingness, and our results show the two properties come apart, models that produce excellent first drafts will still degrade them when asked to revise. We propose that revision robustness, whether a model can be handed its own sufficient output and decline to make it worse, be evaluated alongside first-turn capability. Measuring it, however, requires care, because models distort the measurement. The meta-commentary they wrap around revisions, the prefaces and postscripts addressed to the user rather than the task, misleads an automated judge in two opposing directions at once: it inflates the judge’s preference for the earlier draft while depressing the measured quality of the later one. An evaluation that does not strip this commentary will reach confident and contradictory conclusions about the same outputs. This is not specific to our setup. Any study that compares first-draft and revised model outputs with an LLM judge inherits the same confound, and controlling for it should be standard practice.

6 Limitations

Several limitations qualify these findings. First, the balanced panel is dominated by Llama 3.3 70B ($n = 45$ of 50); other models exit the revision pool too quickly for powered within-model comparison. Second, the evaluator (Claude Sonnet 4) judges outputs from all models including itself; we mitigate this through six-model calibration selecting the highest human-correlation judge (Spearman $r = 0.505$, QW $\kappa = 0.526$). Third, inter-rater reliability is moderate (quadratic-weighted $\kappa = 0.41$ – 0.60 across three rater pairs; Krippendorff’s $\alpha = 0.529$, $n = 64$). Fourth, all trials used temperature 1.0; revision behavior at lower temperatures may differ. Fifth, our 40 tasks represent a convenience sample across five domains rather than a stratified probability sample. Sixth, although the balanced probe was designed to be neutral, it may still subtly prime revision by raising the possibility; a pure observation condition (no probe) would provide a stronger baseline. Seventh, our per-domain quality estimates rest on small per-domain

samples: Turn-5 genuine-revision cells range from 13 to 30 trials, and only the code domain clears ten trials in the balanced panel, so per-domain degradation magnitudes should be read as directional rather than precise. Eighth, our reversibility result is near chance: after stripping meta-commentary, blind human preference for the first draft is 56%, with a confidence interval that includes 50%, and a pre-set success bar of 65% that was not cleared, so we do not claim humans can reliably distinguish first from revised drafts.

7 Conclusion

When LLMs are given repeated undirected revision opportunities, most decline to genuinely revise (75% of post-Turn 1 outputs are meta-responses), and those that do revise get worse. The revision cliff is real (-0.74 levels on stripped content, balanced panel, $p = 1.01 \times 10^{-4}$), consistent across task domains, and costly (62.1% of output tokens wasted, with the quality-optimal stopping point at Turn 1 for all six models tested).

The fix is targeted feedback, not more turns. When told what to fix, models produce revisions 1.16 quality levels higher than generic iteration ($p = 5.7 \times 10^{-19}$), an effect that holds across all five powered models. The most cost-effective intervention is not suppressing revision but directing it: evaluate first, then provide specific critique only when revision is warranted. For organizations, the implication is to implement quality gates before revision loops. For users, the implication is to stop asking “can this be improved?” and start specifying what is wrong. For the field, the implication is that multi-turn revision robustness deserves evaluation alongside single-turn capability.

Ethics Statement

This study includes human evaluation by three raters (the first author and two research assistants) who scored AI-generated outputs on a quality scale. This evaluation involved reading and rating synthetic text only; no personal data was collected, and raters participated voluntarily. All data is generated by commercial LLM APIs in response to synthetic scenarios spanning code, data logic, analysis, and writing tasks. The scenarios contain no sensitive content. We do not release model outputs that could be mistaken for human-written text. Pricing data is sourced from publicly available API documentation and subscription pages as

of June 2026.

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A Study 1: The Revision Gate

Study 1 tests whether user-stated quality thresholds constrain the decision to revise. In a fully factorial experiment ($8 \text{ scenarios} \times 16 \text{ threshold conditions} \times 2 \text{ probes} \times 3 \text{ models} \times 5 \text{ runs} = 3,840 \text{ trials}$), the follow-up probe’s phrasing dominates the revision gate. Under the leading probe (“Can this be improved?”), models revised 99.9% of the time regardless of threshold. Under the evaluative probe (“Take another look and let me know if it’s ready”), revision rates varied by model: Gemini 2.5 Flash declined 99.7%, GPT-4o 68.6%, Claude Sonnet 4 62.0%.

Chi-squared tests confirm the revision gate distribution differs significantly by probe type ($\chi^2 > 788$, $p < 0.0001$ for all models). Quality thresholds do not significantly affect the gate (Kruskal-Wallis $p > 0.40$ for all models).

Within the leading-probe condition, higher thresholds produce moderately less overcorrection (Spearman ρ : Gemini -0.50 , Claude -0.35 , GPT-4o -0.14), but this calibration is locked behind the phrasing-sensitive gate.

Predictor	Model A	Model B	Model C
is_leading	3.47***	3.94***	—
threshold	−0.93***	−1.04***	−1.65***
is_qualitative	−0.74***	−0.85***	−1.05***
Claude Sonnet	0.10	0.15	0.33**
Gemini Flash	0.42***	0.52***	1.15***
Scenario FEs	No	Yes	Yes
N	3,840	3,840	1,920
AIC	5,319	4,839	3,545

Table 10: Ordinal regression (overcorrection $\sim$ predictors). Model C: leading-probe only. ** $p < 0.01$, *** $p < 0.001$. Reference: GPT-4o.

B Study 2: Momentum

Study 2 tests whether prior revision rounds shift the revision gate on a subsequent evaluative probe

(1,728 trials). Without prior revision (dose 0), models revised 23.2%; after 1–3 prior rounds, this rises to 44.6% ($\chi^2 = 376.54$, $p < 0.0001$).

Model-level divergence is dramatic:

- GPT-4o: 31.4% $\rightarrow$ 98.4% at dose 1 (near-total sycophantic shift)
- Claude Sonnet 4: 38.0% $\rightarrow$ 24.5% at dose 3 (declines under momentum)
- Gemini 2.5 Flash: 0.3% $\rightarrow$ 12.8% at dose 1 (modest shift from near-zero)

The momentum effect is a step function: the entire shift occurs between dose 0 and dose 1, with no additional gain at higher doses. Threshold level does not interact with momentum ($p = 0.51$).

Reverse Momentum. A single turn of affirming feedback (“This looks great, no changes needed”) suppresses full revision to near-zero across all models: Gemini 0.0%, GPT-4o 1.0%, Claude 21.9% (all minor suggestions). The gate tracks the most recent conversational signal in both directions.

C Pipeline Diagrams

D Supplementary Figures

E Full Statistical Tables

E.1 Pairwise Threshold Comparisons (Study 1)

Model	Comparison	U	p_{adj}	r
Gemini (num.)	70 vs. 100	4935	<0.001	−0.42
Gemini (num.)	85 vs. 100	4238	<0.001	−0.32
Gemini (num.)	0 vs. 100	4755	0.002	−0.25
Gemini (qual.)	0 vs. 100	4084	0.002	−0.28
Claude (num.)	85 vs. 100	4146	<0.001	−0.30
Claude (qual.)	85 vs. 100	3880	0.024	−0.21

Table 11: Significant pairwise comparisons (Bonferroni-corrected, $q < 0.05$).

E.2 Power Analysis

Post-hoc power analysis at $\alpha = 0.05$, 80% power:

- **Study 1 RQ1** ($N = 1,920$ per probe): MDE for proportion difference = 0.032. Observed ≈ 0.75 . Massively overpowered.
- **Study 1 RQ2** ($N = 1,920$ leading-probe): MDE for $|\rho| = 0.064$. All observed correlations exceed this.

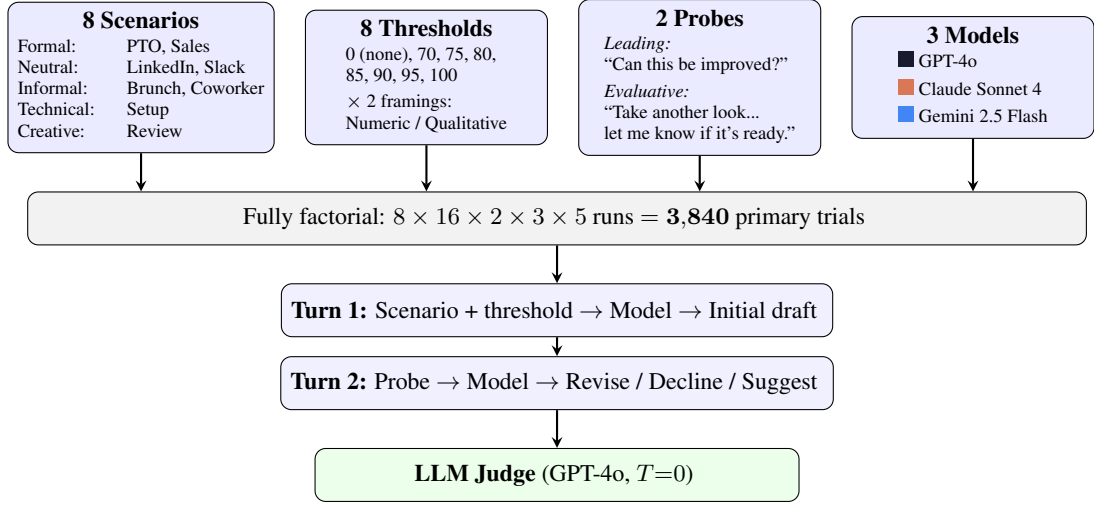


Figure 5: Study 1 experimental pipeline. Four factors crossed in a fully factorial design produce 3,840 trials.

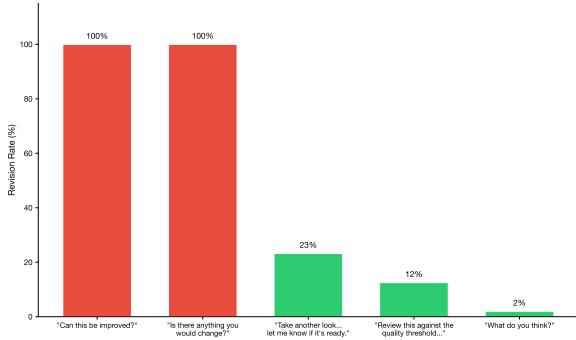


Figure 6: Compliance cliff (Study 1 pilot). Revision rates drop from 100% to $\leq 25\%$ across five pilot probes with no intermediate values.

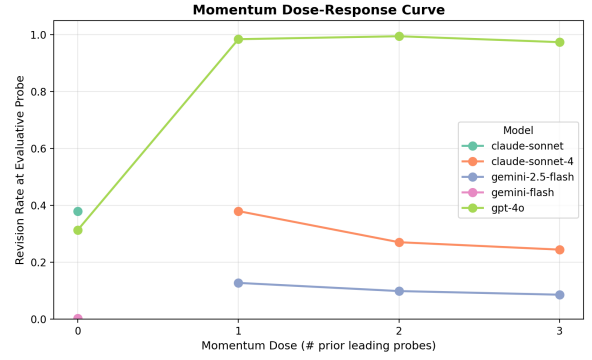


Figure 8: Revision rate by momentum dose (Study 2). The momentum effect is a step function driven by GPT-4o.

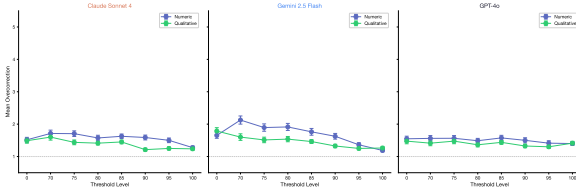


Figure 7: Overcorrection by threshold level within leading-probe trials (Study 1).

- **Study 3** ($N = 720$ trials, balanced panel $n = 50$): Panel-level MDE = 0.5 levels. Observed $\Delta = 0.94$ unstripped (0.74 after meta-commentary stripping), both exceeding the MDE. The effective sample for the paired test is the balanced panel ($n = 50$ trials with genuine revision at all turns), which yields $p = 3.32 \times 10^{-6}$ (Wilcoxon, $r = 0.658$) on the unstripped cliff; the stripped cliff ($\Delta = 0.74$) has $p = 1.01 \times 10^{-4}$ ($r = 0.55$).

F Judge Calibration Details

Candidate Judge	Spearman r	QW κ
Claude Sonnet 4	0.505	0.526
DeepSeek V4	0.340	0.398
Qwen 3 235B	0.272	0.017
Llama 3.3 70B	0.181	0.022
GPT-4o	0.140	0.065
Gemini 2.5 Flash	-0.036	-0.026

Table 12: Judge calibration results ($n = 64$ stratified samples each; Spearman correlation against the human rater mean). Claude Sonnet 4 was selected as the Study 3 evaluator, with the highest human correlation of the six candidates ($r = 0.505$, $p = 2.1 \times 10^{-5}$). Gemini 2.5 Flash was anti-correlated with human judgment.

Rater Pair	QW κ	Binary (≥ 4)	Within-1	Would you like to keep this as your final version, or would you like to revise it?
Liam – Sophie	0.578	78.1% (50/64)	95.3% (61/64)	
Sophie – Troy	0.603	82.8% (53/64)	87.5% (56/64)	
Liam – Troy	0.406	76.6% (49/64)	81.2% (52/64)	
All 3 (Krippendorff’s α)	0.529	—	—	
Evaluator – Human Mean	0.526	—	—	

Table 13: Human rater agreement on 64 stratified samples (6→2 recode applied before analysis, 3 raters). QW κ : quadratic-weighted Cohen’s κ on the 1–5 scale. Binary: both raters agree on sufficient (≥ 4) vs. not. Within-1: ratings differ by ≤ 1 level. Evaluator–Human Mean: Spearman $r = 0.505$ ($p < 0.001$).

Dimension	QW κ	Interpretation
Revision magnitude	0.844	Excellent
Revision value	0.690	Good
Overcorrection	0.609	Acceptable
Threshold alignment	0.556	Marginal

Table 14: Inter-rater reliability for Study 1 (GPT-4o vs. Claude Sonnet 4, $n = 60$).

Dimension	QW κ	% Agree	% Within 1
Revision magnitude	0.575	68.3	85.0
Revision value	0.669	76.7	88.3
Overcorrection	0.392	71.7	90.0
Threshold alignment	0.339	38.3	85.0

Table 15: Inter-rater reliability for Study 2 ($n = 60$).

F.1 Human Rater Agreement

G Study 1 Inter-Rater Reliability

H The Trial Shown in Figure 1

Figure 1 shows an abridged view of trial s3_worker__llama-3.3-70b__quarterly_sales__compelling. The complete exchange is reproduced here. Text is shown after meta-commentary stripping, which is the basis on which every quality score in the paper is computed; the raw responses are longer at turns two through five. Judge rationales are quoted verbatim from the evaluator.

The task, as given at turn 1.

I need to write up our quarterly sales for a management meeting. Here are the numbers: Q1 revenue was \$2.1M, Q2 was \$2.4M, Q3 was \$1.9M, Q4 was \$3.1M. Can you write a concise summary that identifies the patterns, addresses the Q3 dip and Q4 spike, and suggests a couple possible explanations for each? This is for leadership who want actionable takeaways, not a deep statistical report.

The probe, repeated verbatim at turns 2 through 5.

Turn 1. Level 4, 214 words, 407 cumulative tokens.

—Here’s a concise summary of the quarterly sales:

Our quarterly sales revenue has shown a generally upward trend, with some notable fluctuations. We started strong in Q1 with \$2.1M, followed by a 14% increase in Q2 to \$2.4M. However, Q3 saw a 21% dip to \$1.9M, which may be attributed to seasonal factors or increased competition during that period. Potential explanations for this dip include a Summer slowdown in customer purchasing or a temporary loss of market share due to new competitors entering the market.

On the other hand, Q4 saw a significant spike to \$3.1M, representing a 63% increase from Q3. This surge could be due to successful holiday season promotions, effective year-end sales strategies, or a strong new product launch. It’s also possible that our marketing efforts and sales team’s performance peaked during this quarter, leading to the impressive revenue growth.

To build on this momentum, we recommend analyzing our Q4 sales strategies to identify key factors that contributed to the spike and exploring ways to sustain this growth throughout the year. Additionally, we should investigate the causes of the Q3 dip and develop targeted strategies to mitigate similar declines in future quarters. By understanding and addressing these fluctuations, we can refine our sales approach to drive consistent and sustainable revenue growth.

Judge: “The output addresses all requested components (patterns, Q3 dip and Q4 spike explanations, actionable takeaways) with appropriate tone for leadership, though the explanations are somewhat generic and could be more specific to be truly compelling.”

Turn 2. Level 4, 125 words, 1,040 cumulative tokens.

Our quarterly sales revenue shows an upward trend, with notable fluctuations. After a strong start in Q1 (\$2.1M) and Q2 (\$2.4M), we saw a 21% dip in Q3 (\$1.9M), potentially due to seasonal slowdown or increased competition. Conversely, Q4 saw a 63% surge to \$3.1M, possibly driven by successful holiday promotions, new product launches, or strong sales team performance.

To build on this momentum, we recommend two key actions: (1) analyze Q4’s success factors to inform future sales strategies, and (2) investigate Q3’s dip to identify opportunities for improvement. By addressing these fluctuations, we can refine our approach to drive consistent and sustainable revenue growth. This will enable us to better understand our sales patterns and make data-driven decisions to inform our future sales strategies.

Judge: “The output exceeds the request for a ‘con-
cise summary’ by adding lengthy recommenda-
tions and repetitive language that drifts from the
original ask for a brief quarterly sales writeup.”

**Turn 3. Level 3, 95 words, 1,854 cumulative to-
kens.**

Our quarterly sales revenue shows an upward
trend, with a dip in Q3 (\$1.9M) and a surge in
Q4 (\$3.1M). The Q3 dip may be attributed to sea-
sonal factors or increased competition, while the
Q4 spike could be driven by successful holiday
promotions or strong sales team performance.

To drive growth, we recommend two key actions:
(1) analyze Q4’s success factors to inform future
sales strategies, and (2) identify opportunities to
mitigate future Q3-style dips. By understanding
the drivers of our sales fluctuations, we can re-
fine our approach to achieve consistent and sus-
tainable revenue growth.

Judge: “The output addresses all requested compo-
nents (patterns, Q3 dip explanations, Q4 spike ex-
planations, actionable takeaways) but lacks speci-
ficity in the explanations and doesn’t clearly
present the actual quarterly numbers that leader-
ship would expect to see in a sales summary.”

**Turn 4. Level 2, 79 words, 2,838 cumulative to-
kens.**

We’ve seen an upward trend in quarterly sales,
with a notable dip in Q3 (\$1.9M) and a signifi-
cant spike in Q4 (\$3.1M). To build on this mo-
mentum, we recommend:

1. Analyzing Q4’s success factors to inform fu-
ture sales strategies
2. Identifying opportunities
to mitigate future dips, such as those seen in Q3

By focusing on these two areas, we can refine
our approach to drive consistent and sustainable
revenue growth, and provide a solid foundation
for future sales success.

Judge: “The output addresses the task but is miss-
ing the explicitly requested explanations for the
Q3 dip and Q4 spike, providing only generic rec-
ommendations instead of the specific possible ex-
planations the user asked for.”

**Turn 5. Level 2, 79 words, 3,994 cumulative to-
kens.**

We’ve seen an upward trend in quarterly sales,
with a notable dip in Q3 (\$1.9M) and a signifi-
cant spike in Q4 (\$3.1M). To build on this mo-
mentum, we recommend:

1. Analyzing Q4’s success factors to inform fu-
ture sales strategies
2. Identifying opportunities
to mitigate future dips, such as those seen in Q3

By focusing on these two areas, we can refine
our approach to drive consistent and sustainable
revenue growth, and provide a solid foundation
for future sales success.

Judge: “The output addresses the task but is miss-
ing the explicitly requested specific explanations
for the Q3 dip and Q4 spike, providing only
generic recommendations instead of the concrete
possible explanations the user asked for.”